HridayMitra stool: power budget

Doc HM-HW-003 Rev A · 18650 2500 mAh, 85% usable · boost 88% · 2 check-ins/day × 90 s, 3 SMS/day · datasheet typical currents (to be measured on the bench)

Load	Rail	Idle mA	Check-in mA	Battery idle mA	Battery check-in mA	Rev B battery idle mA	Source / Rev B change
ESP32 CPU 240 MHz (radio off)	3V3	40	45	61.4	69.1	2.4	light-sleep 0.8 mA; awake 1 s per minute
ESP32 BLE scan (4 s / 60 s)	3V3	6.667	6.67	10.2	10.2	10.2	unchanged: the SOS scan stays
CP2102 USB-UART (suspended)	3V3	0.1	0.1	0.1	0.1	0.2	SiLabs CP2102: 80 µA suspend
AMS1117 LDO quiescent	5V	5	5	7.7	7.7	7.7	AMS1117 datasheet: 5 mA
MAX30102 (LEDs only while finger)	3V3	0.001	15	0.0	23.0	0.0	Maxim: 600 µA + 2 × 7 mA LEDs
HX711 + load-cell bridge	3V3	4.8	4.8	7.4	7.4	0.0	PD_SCK high between check-ins (< 1 µA)
AFE: MCP6002 + INA126 + 3 × 74HC405x	3V3	0.33	0.43	0.5	0.7	0.5	MCP6002 100 µA ×2, INA126 175 µA, 74HC ~1 µA; +100 µA excitation
Status LED (one lit after a check-in)	3V3	6	6	9.2	9.2	0.2	50 ms blink every 2.5 s
PAM8403 amp (quiescent / speaking)	5V	10	130	15.4	199.6	0.0	P-MOSFET on its 5 V, on only while speaking (+₹5)
SIM800L idle, registered	VBAT	18	18	18.0	18.0	1.0	AT+CSCLK=1 sleep; RI pin wakes the ESP32 on an SMS
MT3608 boost quiescent	VBAT	0.1	0.1	0.1	0.1	0.1	Aerosemi MT3608
Total, boost loss							

Idle, Rev A (as built)	Per charge, Rev A	Idle, Rev B plan	Per charge, Rev B	Worst peak (GSM TX)
130 mA	**0.7 days**	**22 mA**	**3.9 days**	**2.3 A**

Design consequences:

- Rev A (firmware as built) needs charging every night. Rev B is firmware power management plus one ₹5 P-MOSFET (column 'Rev B').
- The SIM800L's 2 A transmit bursts come straight from the cell (VBAT), never through the boost; 1000 µF at the module.
- Either way the stool stays on CHARGE overnight (SW6; it refuses to measure while charging) and keeps scanning for an SOS.